PrepLang: a data-aware compiler for data-preparation pipelines that reuses its proofs on new batches

Nikhil V[a,*], Ranjithkumar S[b]

[a] *School of Computer Science and Engineering, Vellore Institute of Technology, Vellore 632014, Tamil Nadu, India*
[b] *Department of Analytics, School of Computer Science and Engineering, Vellore Institute of Technology, Vellore 632014, Tamil Nadu, India*

Abstract

PrepLang is a compiler that checks a data-preparation pipeline against its first real input. It reads every CSV cell, the Parquet footers and the image headers at compile time, proves or refutes the pipeline's data contracts before any row is processed, and removes the checks that this input makes redundant. The generated pandas and OpenCV program keeps each removal behind a guard, so a later batch reruns only the removed code whose facts it breaks. On 426 later months of five public datasets the compiled pipelines matched a fresh compilation every time.

Keywords: domain-specific language, data-aware compilation, abstract interpretation, speculative optimization, deoptimization, data validation

Metadata

1. Motivation and significance

Much of the hidden cost of machine-learning systems sits in their data dependencies and glue code rather than in the models [1]. A large share of that glue is data preparation: a script over pandas [2] or OpenCV [3] that finds out only at run time whether the column it scales holds numbers, or whether the pixels it saves as PNG are floats. Validators such as TFDV [4], Deequ [5], Great Expectations [6] and pandera [7] check the data, yet they never look at the code that transforms it.

*Corresponding author
Email addresses: atnikhil06@gmail.com (Nikhil V), ranjithkumar.s@vit.ac.in (Ranjithkumar S)

Nr.	Code metadata description	Please fill in this column
C1	Current code version	v2.1.0
C2	Permanent link to code/repository used for this code version	`https://github.com/Nikhi00718/preLang/tree/v2.1.0`
C3	Permanent link to Reproducible Capsule	Docker image `ghcr.io/nikhi00718/preplang:2.1.0` (`Dockerfile` in the repository); `eval/reproduce.sh check` builds, tests and regenerates every number of this article from the committed results
C4	Legal Code License	MIT
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	C++17 (no external libraries); Python 3 with pandas, NumPy, PyArrow, OpenCV, matplotlib; Lean 4; CMake
C7	Compilation requirements, operating environments and dependencies	CMake 3.16+, a C++17 compiler (tested in CI: Visual Studio 2022 on Windows, GCC on Ubuntu and macOS); Python 3.12+ with pandas (2.0+, 3.0 tested), PyArrow, OpenCV, matplotlib (charts); Lean 4.34.1 (optional, for the proofs)
C8	If available, link to developer documentation/manual	`README.md` (user manual), `LANGUAGE.md` (reference), `ARCHITECTURE.md`, `CONTRIBUTING.md`; the READMEs of `eval/`, `paper/`, `proofs/`, `docs/`
C9	Support email for questions	atnikhil06@gmail.com

Table 1: Code metadata (mandatory)

Batch preparation offers something a compiler can rarely count on: the input is already on disk when the pipeline is compiled. PrepLang reads it. Where F# type providers infer types from a sample [8], PrepLang takes values from every CSV cell, Parquet footer and image header and carries what it learns from one statement to the next. Each contract is proven or refuted before any row is processed, and checks or operations that the input makes pointless are removed.

A proof about this month's file does not carry over to next month's, even when the layout is unchanged. For each removal, PrepLang therefore records the facts it relied on, the removal's *item certificate*, and the generated program tests them against every new batch. While they hold, the removal stands. When a batch breaks them, the removed code runs again for that batch, much as dynamic compilers fall back from optimized code [9, 10]; we call this *selective deoptimization*. A numeric column may also switch between whole and decimal numbers when compiling with the other type yields the same program. A batch is refused only when no run-time code can recover a fact, for instance when a numeric column arrives as text.

Table 2 sets PrepLang against the closest tools. Validators see the values but check them away from the code, on every batch; static analyses follow the code but see only declared or inferred structure; provenance-based screening [11] runs the pipeline to catch issues such as leakage between training and test data, but keeps nothing for the next batch. We know of no other tool that decides value contracts at each point of a program from the real

Table 2: PrepLang and the closest tools. *Values*: missing values and value ranges.

Tool	Decides at	Knows about the data	Follows code	On a new batch
TFDV [4], Deequ [5], GX [6], pandera [7]	run time	values, from each batch	no	validates it again
F# type providers [8]	compile time	types, from a sample	no	–
Zhuang and Lu [12]	compile time	column types, from the code	yes	–
Gerstl et al. [13]	compile time	R data-frame shapes, optionally from the data	yes	–
Mirji [14]	compile time	declared schemas	yes	checks the output schema
Schelter et al. [11]	development time, by running it	record-level provenance	yes	–
PrepLang	**compile time**	**types and values of every cell; image headers**	**yes**	**admits by certificate; reruns only the removals it breaks**

data and carries those decisions forward to later batches.

2. Software description

2.1. Software architecture

The compiler [15] (`src/`, about 11,800 lines of C++17 with no external libraries) is a conventional pipeline of lexer, LL(1) parser, semantic analysis, IR, optimizer, certifier and code generator (Fig. 1). Its schema readers scan CSV files in parallel and read only the footers of Parquet files and the headers of PNG and JPEG images. The semantic analyzer builds an abstract state of each dataset (types, missing values, value intervals, image properties) and decides every contract against it. The optimizer removes, speculatively, whatever that state shows to have no effect. The certifier then reanalyzes the program under weaker input facts, without touching the data, to find the admission certificate and one item certificate per removal. Finally, the back end emits a single Python program that admits or refuses each batch, guards every removed instruction with `deopt(i)`, and writes the certificates to `preplang_contract.json`.

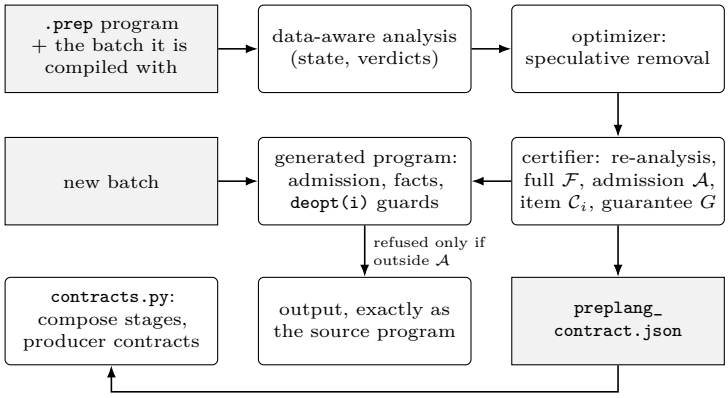


Figure 1: How PrepLang compiles once and reuses the result. The certifier re-runs the analysis with weaker input facts, reading no data; on a new batch, the generated program runs a removed instruction i only if the batch breaks its item certificate $\mathcal{C}_i$.

`proofs/` holds 1477 lines of Lean 4 [16] that prove two results: for any sound and monotone analysis, the generated program agrees with a fresh compilation on every batch it admits; and the analysis of a core of the language is sound and monotone. Joins, groups, dates and images fall outside that core. The differential test of Section 4 covers joins and groups, while dates and images are covered only by unit tests and the examples. `tests/` contains 306 checks in 10 suites.

PrepLang installs with `pip` or runs from the `Dockerfile`; releases ship prebuilt compilers for Windows, Linux and macOS, and a VS Code extension shows the diagnostics on save. Continuous integration builds and tests on all three systems, runs the tests under AddressSanitizer and UndefinedBehaviorSanitizer with a coverage gate, fuzzes the parsers of untrusted input with libFuzzer and checks the Lean proofs. `LANGUAGE.md` documents the language and `ARCHITECTURE.md` maps each phase to its files.

2.2. Software functionalities

- **Compile-time checking.** Misspelled or dropped columns, type errors and broken contracts (`NOT_NULL`, `BETWEEN`, `UNIQUE`, `REFERENCES`, schemas, image channels and sizes, match rates between images and labels) are reported with the line, an offending row and a hint, as are joins on keys of different types. A lookup join expects each key once, while `JOIN ... MANY` keeps every match. Groups (`AGGREGATE`, with eight functions from `COUNT` to `STD`) keep their facts.
- **Data-driven optimization.** Proven checks and operations without effect are removed, unread columns are never loaded, and image pipelines stream one file at a time; rewrites that alter the output need `--approx`.

- **Tools around the compiler.** `tools/contracts.py` checks the guarantees of one compiled stage against the certificates of the next. `frontend/pandas2prep.py` translates the cleaning statements of a pandas script into PrepLang, up to the first statement it cannot express. With `--facts`, a program compiles from differentially private facts released by the data owner (`tools/release_facts.py`) without opening any input; since the generated program checks those facts on the real data, every run is still exact.
- **Inspection.** Each phase can be printed (`--tokens` through `--python`), along with an execution plan (`--explain`) and JSON diagnostics (`--json`). `PLOT` draws histograms and bar, scatter and line charts with matplotlib, marking contract bounds, and `REPORT` writes a web page comparing each column of a run with the compiler's view of it.

3. Illustrative examples

```
LOAD "students.csv" AS students;
EXPECT students.age NUMBER;
EXPECT students.marks BETWEEN 0 AND 100;
EXPECT students.name NOT_NULL;
FILTER students WHERE marks >= 50;
EXPECT students.marks BETWEEN 60 AND 100;
SAVE students AS "clean.csv";
```

Compiled against the eight-row `students.csv` in `examples/tabular/`, PrepLang proves the first three contracts. The fourth stays a run-time check: after the filter, all the compiler knows is that the marks lie in $[50, 95]$. The range and not-null checks are removed speculatively, and `--certificate` lists what the range check depends on, namely marks of at least 0 and at most 100. Given a later batch that contains a mark of 104, the generated program puts the check back and stops exactly where a fresh compilation would:

```
Deoptimized line 3: EXPECT students.marks BETWEEN 0 AND 100:
  marks goes up to 104, above 100
ContractError: line 3: students.marks has values outside 0 to 100
```

A batch that stays within the facts runs with both checks removed. A batch in which `age` holds text is refused, because no generated code can stand in for analysing the column under another type.

Image pipelines get the same treatment, from file headers alone. `examples/images/linked.prep` links images with their labels, checks them, resizes them, scales the pixels to $[0, 1]$ and saves NumPy arrays. Switching the output format to PNG produces this error before a single pixel is decoded:

```
error[type]: cannot save FLOAT32 pixels as PNG:
             NORMALIZE at line 10 scaled them to [0, 1]
   --> examples/images/linked.prep:13:1
13 | SAVE_IMAGES imgs TO "processed/ml_ready/";
   = hint: image files store 8-bit pixels, so these would come out
     almost black; add CONVERT imgs TO NPY; to keep the float
     values
```

The equivalent OpenCV script runs to completion and writes images that are nearly black.

4. Impact

PrepLang is a research prototype, started in a compiler-design course, and has not yet been used outside the authors' own experiments. The evidence for it is the evaluation in `eval/`; `eval/evaluate.py` derives every number below from the raw results.

- **Correctness.** We generated 8,000 random programs over four seeds, with groups, lookup joins and `MANY` joins (1,018 programs). Compiled with and without the optimizer, each produces the same table, value for value, as an independent Python interpreter of the language that shares no code with the compiler. Run again on a second random batch, none of the 1,569 and 4,814 reuses disagrees with the interpreter. In 1,911 of those batches we flipped one numeric column between whole and decimal values: 1,341 ran or stopped as the interpreter did, 570 were refused (57 of them could not be compiled at all), and none was wrong. Earlier runs exposed three optimizer bugs, which had made 109 and 1 runs wrong, and the CSV fuzzer found an out-of-bounds read; each fix has a regression test.
- **Performance.** On a 200,000-row table the optimized program needs 0.81 s instead of 1.32 s after Python start-up; on the smaller inputs of the census below the median speedup is only 1.05× (at most 1.6×). The image pipeline is as fast as a hand-written streaming script and uses 10 MiB (2^{20} bytes), against 922 MiB for a script that loads every image first.
- **Optimization.** Over 452 programs (the examples and the programs lifted from GitHub scripts), the optimizer cuts 14% of the instructions (2,157 to 1,854), the run-time checks from 43 to 8 and 2% of the columns read. It changes 156 programs, and all 452 write identical files with and without it. The lifted scripts carry no contracts, so every removed check comes from the examples or the monthly pipelines.

- **Real programs.** Of 44 data-preparation bugs reported in GitHub issues, PrepLang catches 18 at compile time, 6 of them once the requirement is written down as an `EXPECT`. The lifter translates 378 of 733 public scripts written for the Kaggle Titanic and House Prices data. Their programs match the original script on 4131 of 4158 new batches and stop or refuse on the rest; none is wrong. Among 945 traffic and bike-sharing scripts it translates 85, and these match the script on every one of 3710 later months of real data; for 469 of the 945 the lifter cannot tell which file is read. It leaves `!=` and `~` untranslated, because on a missing value pandas keeps the row and PrepLang drops it.
- **Comparison.** Five monthly pipelines on public data (NYC green taxi, Citi Bike Jersey City, and the UCI traffic, bike-sharing and Beijing air-quality sets) were each compiled on one month and run on the months after it. They match a fresh compilation on 426 of 426 months (100%), including months in which a column switched between whole and decimal numbers, and never produce a different result. The Citi Bike and air-quality pipelines take their bounds from the domain, not from the data. Each validator learned from the compiled month as its documentation recommends and then judged every later month, scored against the same fresh compilation: Great Expectations and pandera reject 96% of the months, and TFDV accepts 10 on which the program then differs. These tools were not built for this decision; the figures show what it costs to use them for it. All 24 mistakes injected into 5 example programs are reported at compile time, whereas the same steps written as a pandas script crash on 20 of them and quietly produce wrong tables for the other 4.
- **Private compilation.** At $\varepsilon = 1$, compiling from released facts keeps every removal in the taxi and traffic pipelines but only 14% in the bike pipeline, where AIM [17] keeps 54%; in 74 of 184 Titanic compilations a released fact proves false and a removal is undone. No output of 64,203 runs differs from a fresh compilation.
- **Overhead.** Compilation takes a median of 125 ms, most of it spent on the extra compilations that decide which numeric columns may change type, and never more than 24 MiB of memory (65 MiB for a million CSV rows). Admitting a new batch by its certificate changes the run time by -1% to $+2\%$, which is within measurement noise.

`eval/reproduce.sh check` (ten minutes, offline) regenerates these numbers from the committed results; `full` (several hours) downloads the inputs and reruns every experiment.

5. Conclusions

PrepLang compiles a preparation pipeline once against real data and reuses its proofs on later batches, rerunning only the removals a batch breaks. Joins match on a single key column, Python is the only target, a numeric column may change type only if no statement combines it with another column, and the mechanized proofs leave out joins, groups, dates and images.

6. Future plans

Joins on several key columns and code generation for targets other than pandas and OpenCV.

CRediT authorship contribution statement

Nikhil V: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Visualization, Writing – original draft. **Ranjithkumar S:** Supervision, Writing – review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The source code, the evaluation scripts and every raw result are in the repository of the metadata table (C2), under the MIT license. The public datasets are downloaded by `eval/reproduce.sh`; the GitHub script corpora are pinned there by URL, commit and hash.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

Generative AI tools assisted with implementation, literature search, evaluation scripts, analysis and drafting of this manuscript. The authors reviewed all text, code and results and take responsibility for them.

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